

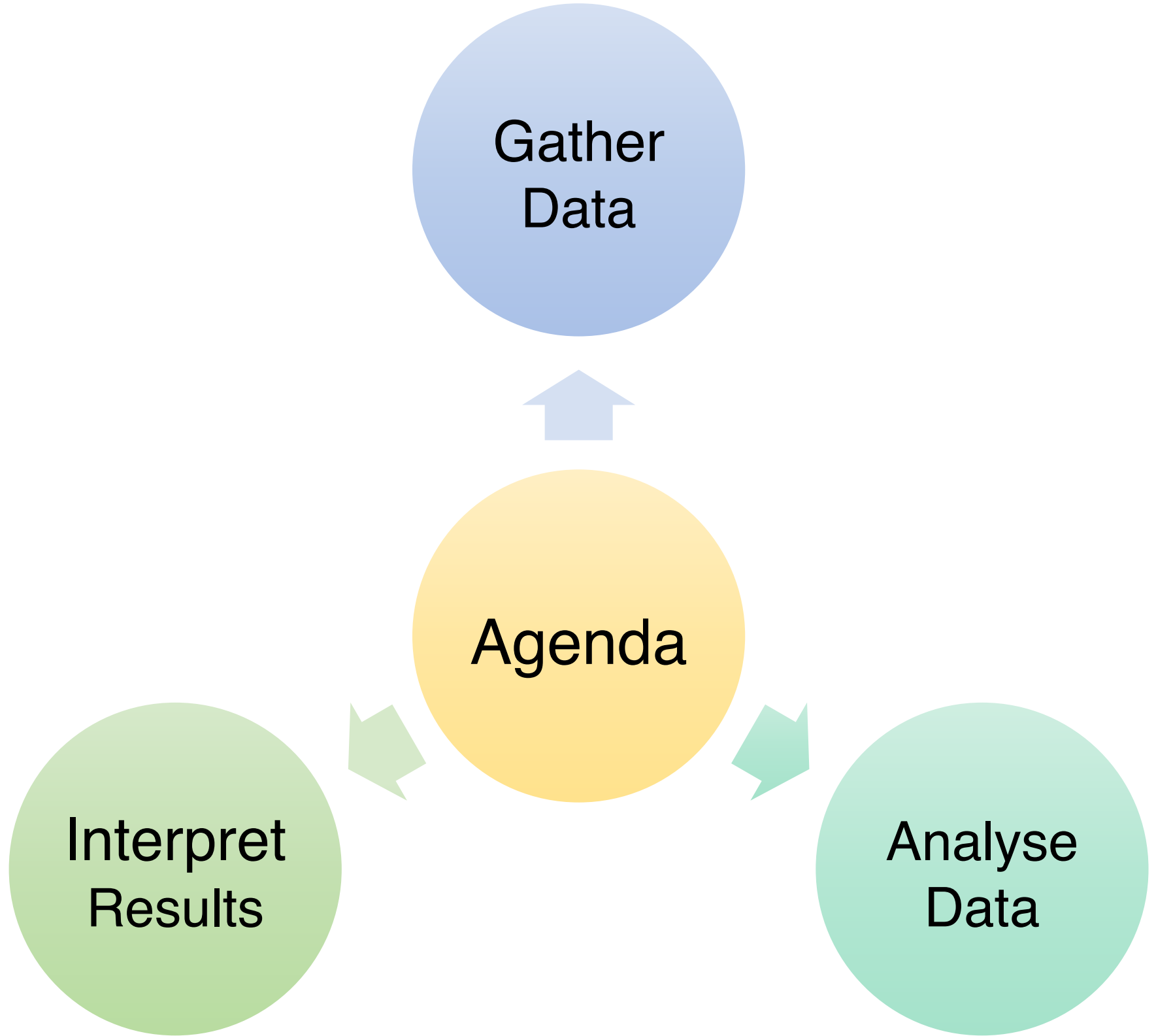
# INFOSYS 704

# **IT Consultancy**

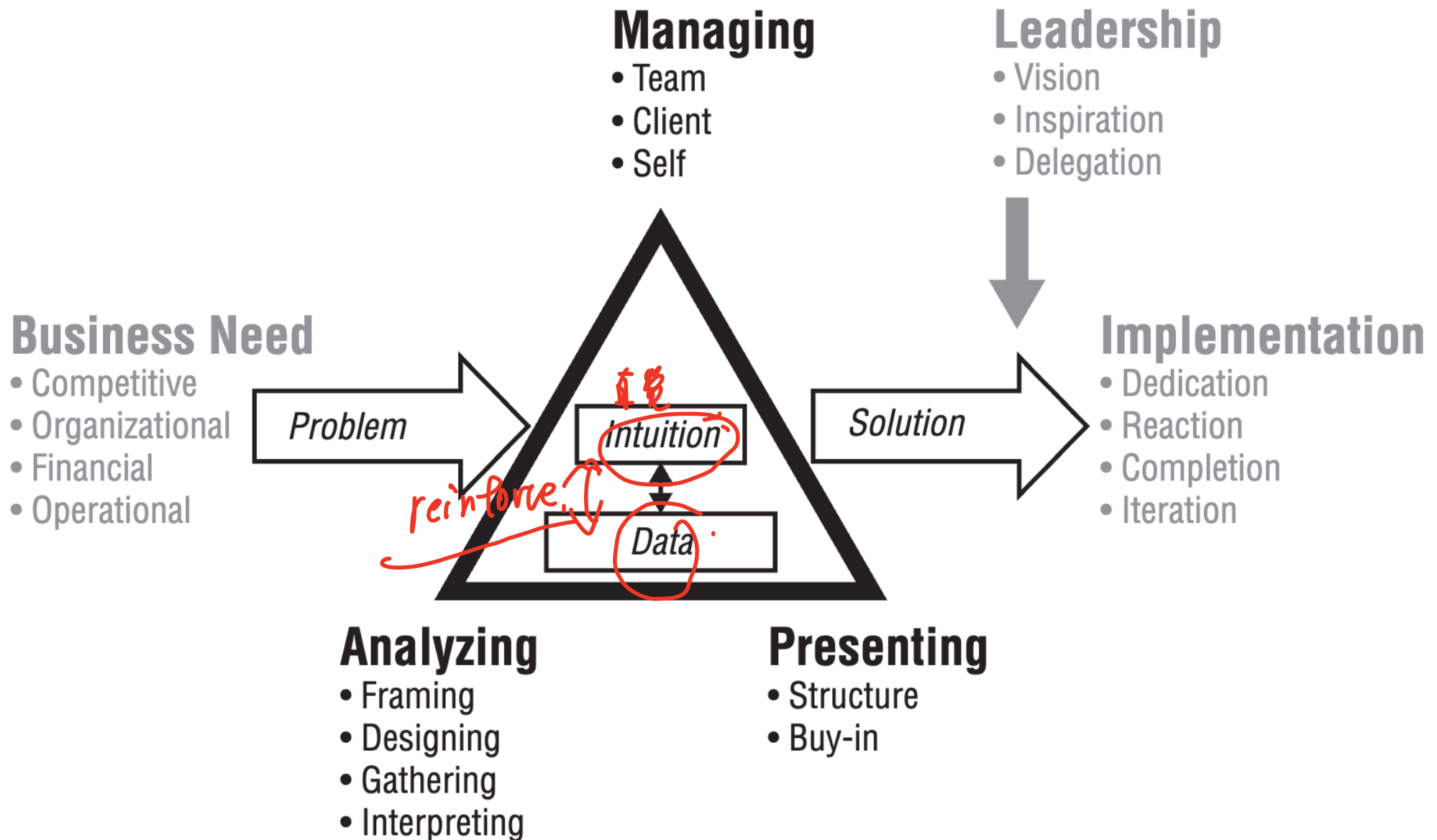
## Lecture 5

**Gathering the Data**  
**Analysing the Data**  
**Interpreting the Results**





# McKinsey Problem Solving Model



Analyze Data

# Gather Data

^  
interview

# Data..data... data... intuition

- By now we have identified the problem, a framework to understand and frame the problem, built issue trees that reflect it, and proposed hypotheses that reflect the problem(s) and issues tree(s).
- Now it is time to collect data to prove or disprove our hypothesis. Ranks amongst the most important consulting skill!
- Balance between facts and intuition
- Research strategies, tools, interviews, and KM

# Research Strategies and Tools

- The problem now is too much data... TMI
- Strategic FOCUS...
- Get the MOST important information as QUICKLY as possible
- Facts are friendly.. Bridge between C & C
- Don't accept "I have no idea".. C & C
- Probe..lack of time.. Insecurity.. Laziness...
- Annual reports, analyst reports, stats, outliers, best practices, bench marking, journals

# Lessons Learned

- Diagnose Data Orientation – facts inhibit effective DM
- Intelligence->Design->Choice->Implementation – Herbert Simon
- Demonstrate power of facts and spread
- Gathering synthesizing and communicating
- Infrastructure to codify studies and expertise inside consulting firm
- Information specialists, conversations with specialists, identify key data sources, Bloomberg

# Some Data Sources

| Category       | Name                            | Description                                                     | Cost | Location                                                                     |
|----------------|---------------------------------|-----------------------------------------------------------------|------|------------------------------------------------------------------------------|
| Search engines | Asianet's Select Search Engines | Over 950 search engines in one place                            | Free | <a href="http://www.asianet.net/search.html">www.asianet.net/search.html</a> |
| Search engines | Findspot                        | Nice search engine guide plus search assistance                 | Free | <a href="http://www.findspot.com">www.findspot.com</a>                       |
| Search engines | Google                          | Easy search that claims access to over 1.3 billion Web pages    | Free | <a href="http://www.google.com">www.google.com</a>                           |
| Search engines | Hotbot                          | Full text of over 100 million Web pages                         | Free | <a href="http://www.hotbot.lycos.com">www.hotbot.lycos.com</a>               |
| Search engines | Alta Vista                      | Power search engine—especially for advanced searches            | Free | <a href="http://www.altavista.com">www.altavista.com</a>                     |
| Search engines | FAST Search                     | Claims access to over 575 million URLs; extensive list of sites | Free | <a href="http://www.alltheweb.com">www.alltheweb.com</a>                     |
| Search engines | Yahoo                           | One of the old standards—some commercialization                 | Free | <a href="http://www.yahoo.com">www.yahoo.com</a>                             |
| Search engines | BPubs.com                       | Allows searches just on business publications                   | Free | <a href="http://www.bpubs.com">www.bpubs.com</a>                             |

# Some Data Sources

| Category            | Name                                | Description                                                                                      | Cost   | Location                                                                                                       |
|---------------------|-------------------------------------|--------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------|
| General information | ABI/Inform Global (Proquest Direct) | Abstracts and some full text for articles in over 1,000 leading journals                         | Varies | Subscription information at <a href="http://www.proquest.com">www.proquest.com</a>                             |
| General information | Academic Universe (Lexis/Nexis)     | General and specific industry and company information; major news wires                          | Varies | Subscription information at <a href="http://www.lexis-nexis.com">www.lexis-nexis.com</a>                       |
| General information | AJR NewsLink                        | Access to over 3,400 U.S. and 2,000 non-U.S. newspapers                                          | Free   | <a href="http://ajr.newslink.org/news.html">ajr.newslink.org/news.html</a>                                     |
| General information | Business & Industry                 | Facts, figures, and key events for international companies                                       | Varies | Subscription information at <a href="http://www.galegroup.com/welcome.html">www.galegroup.com/welcome.html</a> |
| General information | Business Wire                       | Business news and information about industries and companies—latest news                         | Free   | <a href="http://www.businesswire.com">www.businesswire.com</a>                                                 |
| General information | Dow Jones Interactive               | Extensive access to full-text articles from newspapers, magazines, journals, and broadcast media | Varies | Subscription information at <a href="http://askdj.dowjones.com/">http://askdj.dowjones.com/</a>                |
| General information | Individual.com                      | Free company and industry news; can be customized based on your input                            | Free   | <a href="http://www.individual.com">www.individual.com</a>                                                     |

# Existing Client Data

| Type        | Examples                                                   |
|-------------|------------------------------------------------------------|
| Operational | Sales reports, inventory data, call center logs            |
| Financial   | P&L statements, cost breakdowns, balance sheets            |
| Digital/CRM | Web analytics, customer journey data, marketing dashboards |
| HR & Org    | Org charts, employee surveys, performance data             |

# Interviews

- **Stakeholders:**

- Executives (strategic direction)
- Line managers (process knowledge)
- Frontline staff (pain points)
- Customers (needs, satisfaction)

- **Experts**

- Industry experts (through proprietary expert networks)
- Academics or technologists (for technical/innovation topics)
- Regulators or NGOs (in public sector projects)

# Interviews

- Interview guides.. Questionnaires
- Right people.. Gathering data.. Test ideas..  
Increase Buy-in
- Interview guide... sequence of questions...  
pilot it.. Boil down to 3 or 4 MOST important questions
- End with “Is there anything I forgot to ask?”
- Also the power of switching off the recording device..
- Listen and guide interviews
- Send interview guide ahead to interviewees..  
exceptions...post interview follow up

# Interviews

- Rigorous but sensitive...
- Start with less sensitive issues.. Big picture initially.. Slow.. gentle
- Active listening.. Nods.. Facial expressions.. Arms crossing.. Interjections.. Grunts.. also silence.. positive body language..
- Don't allow too many tangents.. hobby horses..
- Recording of interviews.. Online.. Physical..
- Otter translate... Teams/Zoom transcripts
- AI to summarise

# Interview Tips

- Interviewee boss set up meeting
- Interview in pairs
- Listen and don't lead
- Paraphrase
- Indirect approach... triangulation...
- Don't ask for too much.. Simple questions
- Columbo tactic.. Finish.. And then Zing
- Be sensitive.. Give space..
- Difficult interviews... job on the line..
- Always... THANK YOU... note!

# Marketing Process Transform Interview

- **Opening** (5-10 min) Establish context and rapport, Explain objective: improve marketing effectiveness and efficiency, Invite input and questions from interviewee
- **Role, Scope & Priorities** (5 min) Describe role & marketing team structure, Understand top 3 priorities, Identify key KPIs tracked by team
- **Marketing Process Overview** (10-15 min) Map end-to-end campaign process, Explore decision-making and bottlenecks, Identify inefficiencies or unclear workflows
- **Data, Analytics & Measurement** (10 min) Assess how performance & ROI are tracked, Tools & dashboards used, Evaluate access to strategic data
- **Channels & Content** (10 min) Identify used channels & effectiveness, Understand content creation process, Explore opportunities for targeting & personalization
- **Technology & Tools** (5-10 min) Overview of current MarTech stack, Integration with other tools/systems, Pain points & underutilized features
- **Team, Skills & Collaboration** (5-10 min) Assess team capabilities across key skills, Understand collaboration with other functions, Explore enablers and blockers for collaboration
- **Closing** (2-5 min) Thank interviewee, Ask about additional initiatives, Capture final insights or concerns, Zinger!

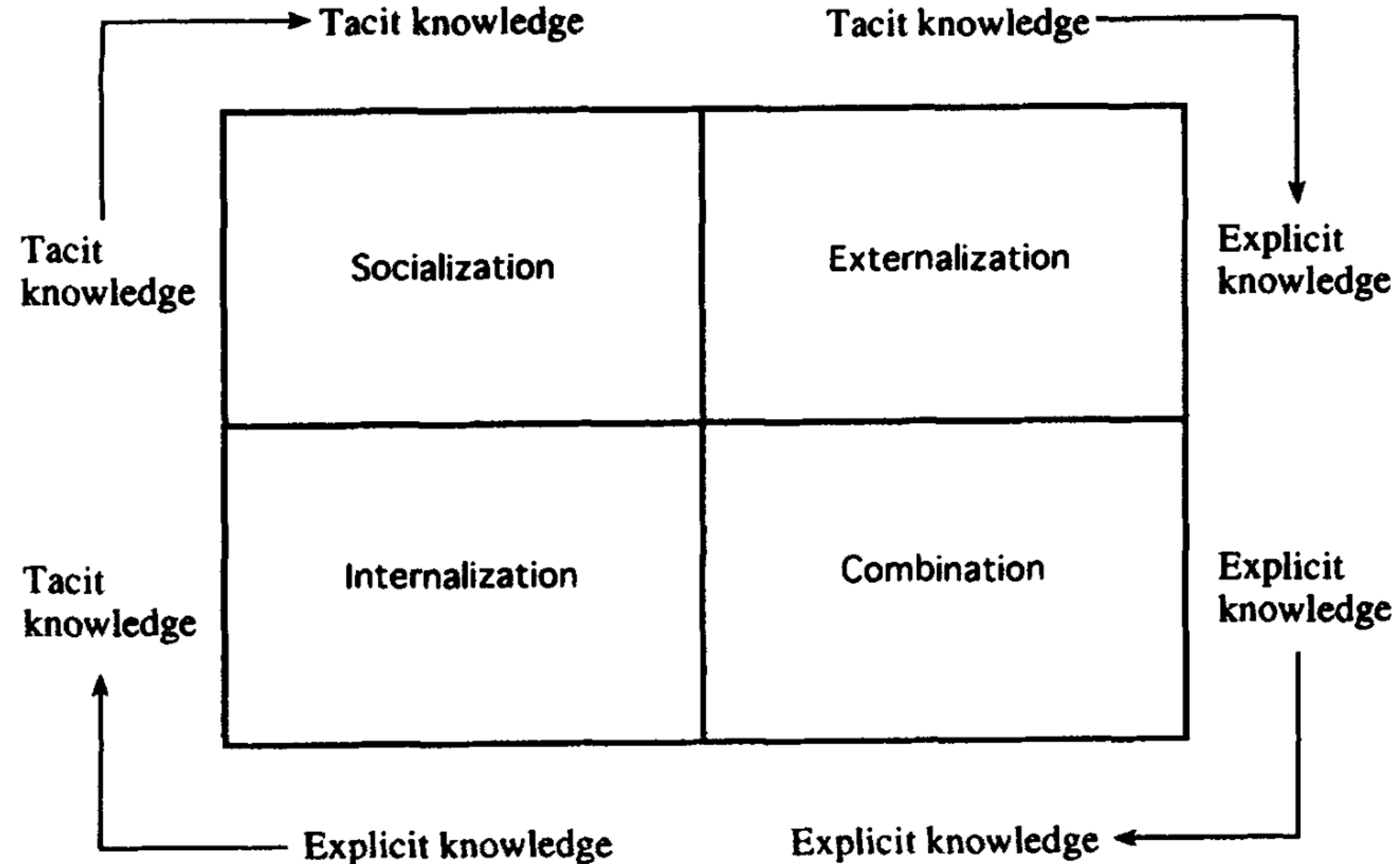
# Digital Transformation Interview

- **Opening** (5-10 min) Set stage for interview & build rapport, Explain project goal: assess readiness and opportunities for digital transformation, Invite open discussion & any key concerns from the interviewee.
- **Role, Responsibilities & Strategic Goals** (5 min) Describe your role and scope within the organization, What are your top priorities for the next 12–24 months?, How does digital transformation align with your strategic objectives?
- **Current Digital Landscape** (10-15 min) What digital tools or platforms are in use today? How do they support or limit your current operations? What gaps or frustrations exist in the current system landscape?
- **Processes & Automation** (10 min) Which processes are already digitized or automated? Where do manual workflows still exist? What areas would benefit most from digital enhancement?
- **Data & Analytics Capabilities** (10 min) How is data currently collected, stored, & used in decision-making? What analytics tools or dashboards are used today? Do you trust the data quality & accessibility?
- **Technology Infrastructure** (5-10 min) How would you describe your current IT architecture? Is it scalable and flexible for future growth? What integrations or legacy system constraints exist?
- **Culture, Skills & Change Readiness** (5-10 min) How ready is your team or function to adopt new digital ways of working? Where are the biggest skill or mindset gaps? What has helped or hindered adoption of past digital initiatives?
- **Closing** (2-5 min) Thank the interviewee for their time and insights Ask if there are any ongoing or planned digital initiatives to note. Open the floor for any final thoughts, concerns, or ideas. Zinger!

# Knowledge Management

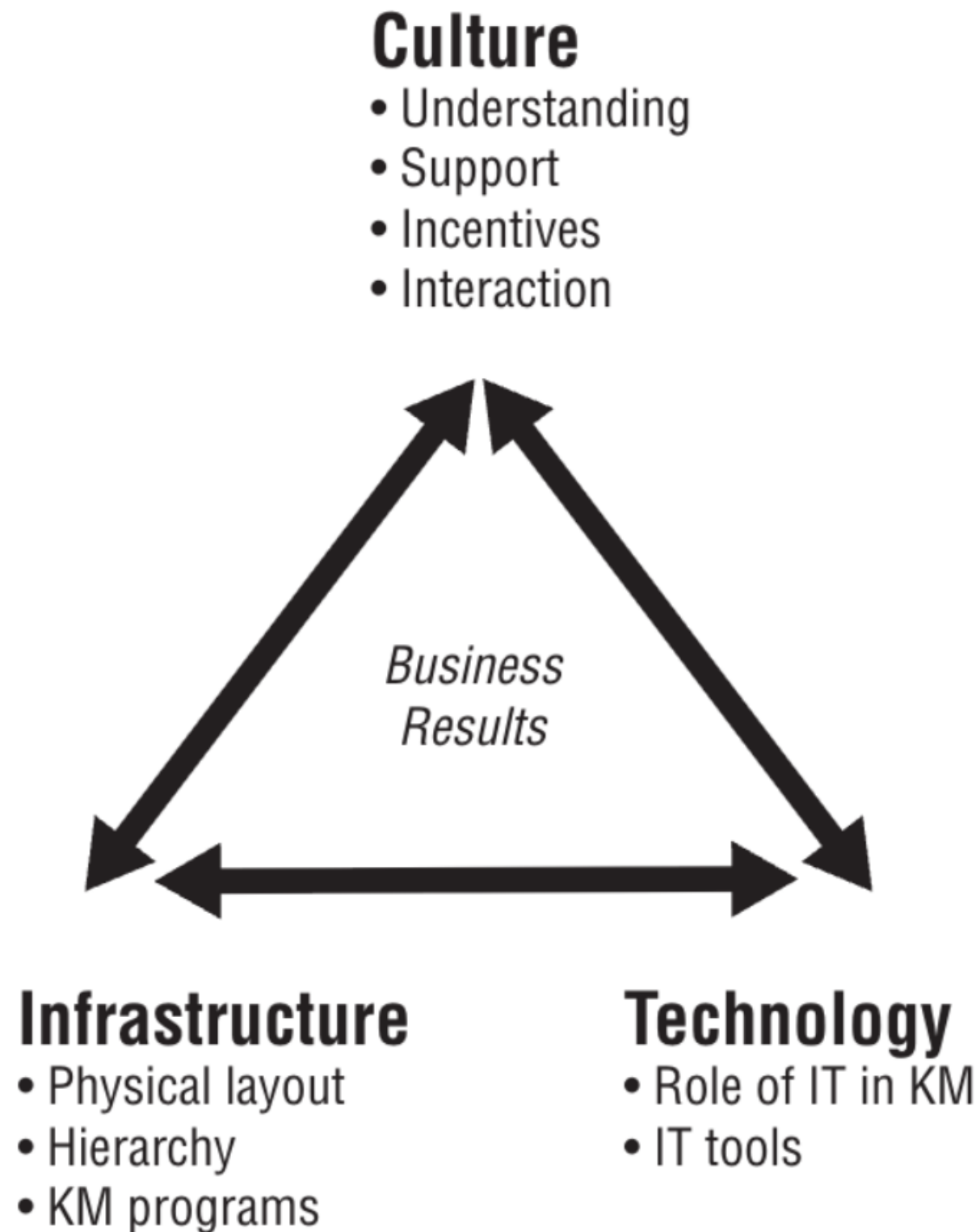
- DIKW
- Mix of information, experience and context in a value adding process
- Uncodified... Codified..
- Codification strategies... platforms
- Don't Reinvent the Wheel..
- Databases... experts.. GenAI
- Know what... know who...

# SECI – Nonaka & Takeuchi 1995



# KM Enablers in R&D

- Rapid Response – from U to C
- Sharing - Practice Olympics
- External K
- GIGO
- KM audit



# Gathering Data

| Method       | Description                           | When Used                |
|--------------|---------------------------------------|--------------------------|
| Client Data  | Internal KPIs, systems, reports       | Always                   |
| Interviews   | With staff, execs, customers          | Early-mid phase          |
| Expert Calls | Industry/tech/market insight          | Complex/niche topics     |
| Benchmarking | Using proprietary or market data      | Competitive assessments  |
| Surveys      | Structured feedback or market insight | Custom research          |
| Fieldwork    | Observations, ride-alongs, audits     | CX, ops, org design      |
| Analytics    | Models, clustering, prediction        | Digital/AI/advanced work |

**Analyse Data**

# Purpose and Process of Analysis

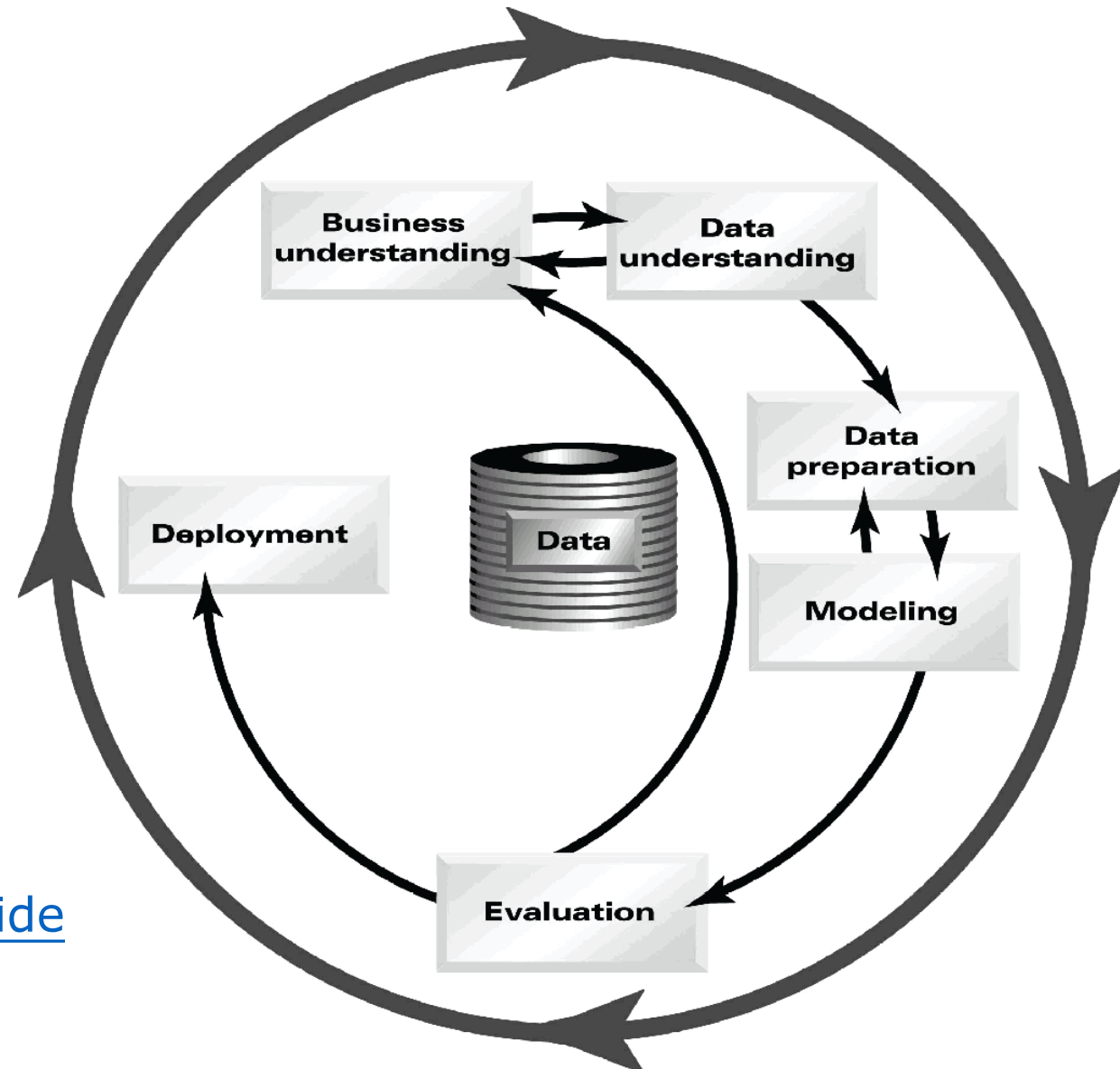
- Descriptive
- Diagnostic
- Predictive
- Prescriptive
- Persuasive

[SPSS Modeller User Guide](#)

[Clementine User Guide](#)

[SPSS Modeller CRISP-DM Guide](#)

[Orange Data Mining](#)



Fayyad, U., Piatetsky-Shapiro, G., & Smyth, P. (1996). From data mining to knowledge discovery in databases. AI magazine, 17(3), 37. <https://www.aaai.org/ojs/index.php/aimagazine/article/viewFile/1230/1131>

Wendler, Tilo, and Sören Gröttrup. Data Mining with SPSS Modeler : Theory, Exercises and Solutions, Springer, 2021. ProQuest Ebook Central, <https://ebookcentral.proquest.com/lib/auckland/detail.action?docID=6631133>.

# Business & Data Understanding

**1. Business and/or Situation understanding** “First is developing an understanding of the application domain and the relevant prior knowledge and identifying the goal of the KDD process from the customer’s viewpoint.” (Fayyad et al., 1996)

- 1.1 Identify the objectives of the business/situation
- 1.2 Assess the situation
- 1.3 Determine data mining objectives, and
- 1.4 Produce a project plan

**2. Data understanding** Data provides the “raw materials” of data mining. This phase addresses the need to understand what your data resources are and the characteristics of those resources. “Second is creating a target data set: selecting a data set, or focusing on a subset of variables or data samples, on which discovery is to be performed.” (Fayyad et al., 1996)

- |                             |                             |
|-----------------------------|-----------------------------|
| • 2.1 Collect initial data  | 2.2 Describe the data       |
| • 2.3 Explore the data, and | 2.4 Verify the data quality |

# Data Preparation & Transformation

**3. Data preparation** After cataloguing your data resources, you will need to prepare your data for mining. “Third is data cleaning and pre-processing. Basic operations include removing noise if appropriate, collecting the necessary information to model or account for noise, deciding on strategies for handling missing data fields, and accounting for time-sequence information and known changes” (Fayyad et al., 1996)

- 3.1 Select the data
- 3.2 Clean the data
- 3.3 Construct the data
- 3.4 Integrate various data sources
- 3.5 Format the data as required

**4. Data transformation** “Fourth is data reduction and projection: finding useful features to represent the data depending on the goal of the task. With dimensionality reduction or transformation methods, the effective number of variables under consideration can be reduced, or invariant representations for the data can be found.” (Fayyad et al., 1996)

- 4.1 Reduce the data
- 4.2 Project the data

# Data Mining Method & Algorithm Selection

**5. Data-mining method(s) selection** “Fifth is matching the goals of the KDD process (step 1) to a particular data-mining method. For example, summarization, classification, regression, clustering, and so on, are described later as well as in Fayyad, Piatetsky-Shapiro, and Smyth (1996).” (Fayyad et al., 1996)

- 5.1 Match and discuss the objectives of data mining (1.1) to data mining methods
- 5.2 Select the appropriate data-mining method(s) based on discussion

**6. Data-mining algorithm(s) selection** “Sixth is exploratory analysis and model and hypothesis selection: choosing the datamining algorithm(s) and selecting method(s) to be used for searching for data patterns. This process includes deciding which models and parameters might be appropriate (for example, models of categorical data are different than models of vectors over the reals) and matching a particular data-mining method with the overall criteria of the KDD process (for example, the end user might be more interested in understanding the model than its predictive capabilities).” (Fayyad et al., 1996)

- 6.1 Conduct exploratory analysis and discuss
- 6.2 Select data-mining algorithms based on discussion
- 6.3 Build/Select appropriate model(s) and choose relevant parameter(s)

# Data Mining

**7. Data Mining** “Seventh is data mining: searching for patterns of interest in a particular representational form or a set of such representations, including classification rules or trees, regression, and clustering. The user can significantly aid the data-mining method by correctly performing the preceding steps.” (Fayyad et al., 1996) This is, of course, the flashy part of data mining, where sophisticated analysis methods are used to extract information from the data.

- 7.1 Create and justify test designs
- 7.2 Conduct data mining – classify, regress, cluster, etc. (models must execute)
- 7.3 Search for patterns

# Interpret Results

# Interpretation

**8. Interpretation** “Eighth is interpreting mined patterns, possibly returning to any of steps 1 through 7 for further iteration. This step can also involve visualization of the extracted patterns and models or visualization of the data given the extracted models.” (Fayyad et al., 1996) We assess and evaluate the models and the results and their reliability. “You are ready to evaluate how the data mining results can help you to achieve your objectives.” (SPSS, 2007)

- 8.1 Study and discuss the mined patterns
- 8.2 Visualize the data, results, models, and patterns
- 8.3 Interpret the results, models, and patterns
- 8.4 Assess and evaluate results, models, and patterns

# Action

**9. Action:** “Ninth is acting on the discovered knowledge: using the knowledge directly, incorporating the knowledge into another system for further action, or simply documenting it and reporting it to interested parties. This process also includes checking for and resolving potential conflicts with previously believed (or extracted) knowledge.” (Fayyad et al., 1996) “Now that you’ve invested all of this effort, it’s time to reap the benefits. This phase focuses on integrating your new knowledge into your everyday business processes to solve your original business problem and/or situation.” (SPSS, 2007)

- 9.1 Discuss how you would apply the knowledge and deploy the implementation
- 9.3 Discuss how you would monitor the implementation
- 9.4 Discuss how you would maintain the implementation
- 9.5 How could you enhance the solution in the future?

# Individual versus Holistic

- Understand the meaning of Individual Analyses. Imagination to put together disparate facts into coherent narrative.
- IF you prove your hypothesis, you go forward.
- If you disprove your hypothesis, then you iterate to revisit and restructure your framework, issues tree, and hypotheses to FIT your data and analysis!
- Data analysis – 80:20/Pareto, Chart / Figure / Table / PPT a Day, Don't make facts fit your solution!
- What's the “So what?” What does all the analysis tell us? Recommendation? Insights? Useful?
- Sanity checks. Bounds of probability. Scenario analysis. Assumptions. Forest vs. Trees.
- Draw inferences. Now intuition takes over from data!

# Adaptation... End Product

- When the Facts Change... Change your Mind
- When facts contradict your hypothesis, change your framework/model/issue tree/hypothesis. Be prepared to kill your babies.
- What is your end product? The actual message you will communicate? What is your story? What does the client need to know to understand your conclusions?
- Make sure your solution fits the client. Strengths Weaknesses Capabilities. See through clients' eyes. CEO focus. What are their priorities. Limits of their abilities.
- What value will your solution add? Payoff? Against time, energy, resources?
- Help clients to make change happen. Put together a compelling narrative,

# Techniques used to Interpret Results

|                       |                                                                                |
|-----------------------|--------------------------------------------------------------------------------|
| <b>Issue Trees</b>    | Break down the problem logically to test hypotheses                            |
| <b>Benchmarking</b>   | Compare KPIs to competitors or industry standards                              |
| <b>MECE Framework</b> | Ensure categories and findings are Mutually Exclusive, Collectively Exhaustive |
| <b>Storylining</b>    | Distill complex findings into structured narratives                            |
| <b>80/20 Analysis</b> | Focus on the few variables driving the most impact                             |

# Customer Churn Analysis

- **Problem:** High churn in telecom unit
- **Data:**
  - Churn rate rose from 14% to 21%
  - Net Promoter Score dropped by 22 points
  - 40% of new customers churned in 90 days
- **Insight:** Network quality and onboarding issues are primary churn drivers
- **Recommendation:** Invest in infrastructure and redesign onboarding

# Marketing ROI Analysis

- **Problem:** Low return on marketing spend
- **Data:**
  - Paid search ROI = 5.3x, 22% of leads
  - Social media ROI = 1.6x, 38% of spend
  - Email ROI = 7.8x, only 6% of spend
- **Insight:** Email outperforms yet is underfunded
- **Recommendation:** Shift budget to email and paid search to increase ROI

# Supply Chain Optimization

- **Problem:** Rising warehouse costs
- **Data:**
  - Labor = 55% of cost, overtime up 20% YoY
  - SKU count ↑ , pick path ↑ 35%
  - 70% of SKUs generate 10% of revenue
- **Insight:** Cost driven by SKU bloat and inefficient layout
- **Recommendation:** Classify SKUs, trim low-value items, optimize layout

